

# Comprehensive Medical Knowledge Base for RAG Testing

## Introduction to Human Health

Human health is influenced by genetics, nutrition, environment, exercise, sleep, mental well-being, and healthcare access. The human body contains multiple interconnected systems including the cardiovascular system, respiratory system, digestive system, endocrine system, nervous system, immune system, and musculoskeletal system.

A healthy adult typically maintains normal blood pressure between 90/60 mmHg and 120/80 mmHg. Resting heart rate for adults generally ranges between 60 and 100 beats per minute. Body temperature is usually around 37 degrees Celsius.

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## Cardiovascular Diseases

### Hypertension

Hypertension, also called high blood pressure, occurs when the force of blood against artery walls remains consistently elevated. Hypertension may not show symptoms initially but can significantly increase the risk of heart disease, stroke, kidney disease, and heart failure.

#### Symptoms

- Headache
- Dizziness
- Blurred vision
- Chest pain
- Fatigue

#### Risk Factors

- Smoking
- Obesity
- High salt intake
- Stress
- Family history
- Physical inactivity

#### Treatment

- ACE inhibitors

- Beta blockers
- Calcium channel blockers
- Lifestyle changes
- Reduced sodium intake

## **Prevention**

- Exercise regularly
  - Reduce alcohol consumption
  - Maintain healthy weight
  - Avoid smoking
  - Monitor blood pressure regularly
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## **Coronary Artery Disease**

Coronary artery disease occurs when coronary arteries become narrowed due to plaque buildup. Reduced blood flow can lead to chest pain or heart attack.

## **Symptoms**

- Chest pain
- Shortness of breath
- Pain in left arm
- Fatigue
- Sweating

## **Diagnostic Methods**

- ECG
- Echocardiogram
- Coronary angiography
- Stress testing

## **Treatment**

- Statins
  - Aspirin
  - Angioplasty
  - Coronary artery bypass surgery
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# Respiratory Diseases

## Asthma

Asthma is a chronic inflammatory disease affecting the airways. Airway narrowing leads to breathing difficulty.

### Symptoms

- Wheezing
- Shortness of breath
- Chest tightness
- Persistent coughing

### Triggers

- Dust
- Smoke
- Pollen
- Exercise
- Cold air

### Treatment

- Inhaled corticosteroids
- Bronchodilators
- Leukotriene modifiers

### Prevention

- Avoid allergens
  - Use inhalers properly
  - Maintain clean environment
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## Pneumonia

Pneumonia is an infection causing inflammation in the air sacs of the lungs. It may be caused by bacteria, viruses, or fungi.

### Symptoms

- Fever
- Cough
- Chills
- Difficulty breathing

- Chest pain

### **Risk Groups**

- Elderly
- Infants
- Immunocompromised individuals

### **Treatment**

- Antibiotics
  - Antiviral medications
  - Oxygen therapy
  - Hydration
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## **Endocrine Disorders**

### **Diabetes Mellitus**

Diabetes mellitus is a chronic disease affecting blood glucose regulation.

#### **Type 1 Diabetes**

An autoimmune condition where the pancreas produces little or no insulin.

#### **Type 2 Diabetes**

Occurs due to insulin resistance and impaired insulin secretion.

### **Symptoms**

- Excessive thirst
- Frequent urination
- Fatigue
- Weight loss
- Blurred vision

### **Complications**

- Neuropathy
- Retinopathy
- Kidney disease
- Cardiovascular disease

## **Treatment**

- Insulin therapy
- Metformin
- Dietary management
- Exercise

## **Prevention**

- Weight management
  - Healthy eating
  - Physical activity
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# **Hypothyroidism**

Hypothyroidism occurs when the thyroid gland does not produce enough thyroid hormone.

## **Symptoms**

- Fatigue
- Weight gain
- Cold intolerance
- Dry skin
- Depression

## **Treatment**

- Levothyroxine
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# **Neurological Disorders**

## **Migraine**

Migraine is a neurological disorder causing severe headaches.

## **Symptoms**

- Pulsating headache
- Sensitivity to light
- Nausea
- Vomiting

## Triggers

- Stress
- Hormonal changes
- Certain foods
- Lack of sleep

## Treatment

- Pain relievers
  - Triptans
  - Preventive medications
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## Epilepsy

Epilepsy is characterized by recurrent seizures caused by abnormal brain activity.

## Symptoms

- Temporary confusion
- Staring spells
- Loss of consciousness
- Muscle spasms

## Treatment

- Anti-seizure medications
  - Surgery in severe cases
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# Gastrointestinal Disorders

## Gastritis

Gastritis refers to inflammation of the stomach lining.

## Causes

- Helicobacter pylori infection
- Alcohol
- Stress
- NSAID overuse

## Symptoms

- Nausea
- Vomiting
- Upper abdominal pain
- Bloating

## Treatment

- Antacids
  - Proton pump inhibitors
  - Antibiotics
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# Peptic Ulcer Disease

Peptic ulcers are sores developing in the stomach or upper small intestine lining.

## Causes

- H. pylori infection
- NSAID use

## Symptoms

- Burning stomach pain
- Bloating
- Heartburn
- Nausea

## Treatment

- Antibiotics
  - Acid suppression therapy
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# Infectious Diseases

## Tuberculosis

Tuberculosis is a bacterial infection caused by Mycobacterium tuberculosis.

## Symptoms

- Chronic cough
- Fever

- Night sweats
- Weight loss

### **Transmission**

Airborne droplets.

### **Treatment**

- Isoniazid
  - Rifampin
  - Ethambutol
  - Pyrazinamide
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## **COVID-19**

COVID-19 is caused by the SARS-CoV-2 virus.

### **Symptoms**

- Fever
- Dry cough
- Fatigue
- Loss of taste
- Breathing difficulty

### **Prevention**

- Vaccination
- Hand hygiene
- Mask usage

### **Complications**

- Pneumonia
  - ARDS
  - Organ failure
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## **Musculoskeletal Disorders**

### **Osteoarthritis**

Osteoarthritis involves degeneration of joint cartilage.



## Symptoms

- Joint pain
- Stiffness
- Reduced flexibility

## Risk Factors

- Aging
- Obesity
- Joint injury

## Treatment

- Pain medications
  - Physical therapy
  - Joint replacement surgery
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# Osteoporosis

Osteoporosis weakens bones and increases fracture risk.

## Symptoms

- Back pain
- Height loss
- Fragile bones

## Prevention

- Calcium intake
  - Vitamin D
  - Weight-bearing exercise
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# Mental Health Disorders

## Depression

Depression is a mood disorder affecting emotional well-being.

## Symptoms

- Persistent sadness
- Loss of interest

- Fatigue
- Sleep disturbances

### **Treatment**

- Psychotherapy
  - Antidepressants
  - Lifestyle modification
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## **Anxiety Disorder**

Anxiety disorders involve excessive fear and nervousness.

### **Symptoms**

- Rapid heartbeat
- Sweating
- Restlessness
- Insomnia

### **Treatment**

- Cognitive behavioral therapy
  - Relaxation techniques
  - Medications
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## **Nutrition and Diet**

A balanced diet includes carbohydrates, proteins, fats, vitamins, minerals, and water.

### **Important Nutrients**

- Iron supports hemoglobin production.
- Calcium supports bone health.
- Vitamin C improves immunity.
- Protein supports muscle repair.

### **Healthy Practices**

- Consume fruits and vegetables.
  - Limit processed foods.
  - Drink adequate water.
  - Reduce sugar intake.
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# Emergency Medicine

## Stroke

A stroke occurs when blood supply to the brain is interrupted.

### Symptoms

- Facial drooping
- Arm weakness
- Speech difficulty

### Emergency Response

Immediate medical care is necessary.

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## Heart Attack

A heart attack occurs when blood flow to part of the heart is blocked.

### Symptoms

- Chest pain
- Sweating
- Nausea
- Shortness of breath

### Emergency Action

Call emergency medical services immediately.

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# Pharmacology Basics

## Antibiotics

Antibiotics treat bacterial infections.

Examples:

- Amoxicillin
- Ciprofloxacin
- Azithromycin

Antibiotics are ineffective against viral infections.

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## **Analgesics**

Analgesics reduce pain.

Examples:

- Paracetamol
- Ibuprofen

Overuse of NSAIDs may damage the stomach lining.

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## **Medical Imaging**

### **MRI**

Magnetic Resonance Imaging uses magnetic fields to create detailed internal images.

#### **Common Uses**

- Brain imaging
  - Spine evaluation
  - Joint injuries
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### **CT Scan**

Computed tomography combines X-rays with computer technology.

#### **Uses**

- Trauma assessment
  - Tumor detection
  - Lung imaging
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# Laboratory Tests

## Complete Blood Count (CBC)

CBC measures:

- Red blood cells
- White blood cells
- Hemoglobin
- Platelets

### Uses

- Detect infections
  - Detect anemia
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## Blood Glucose Test

Used to diagnose diabetes and monitor blood sugar levels.

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# Immunology

The immune system protects the body against pathogens.

### Types of Immunity

- Innate immunity
- Adaptive immunity

### White Blood Cells

- Neutrophils
- Lymphocytes
- Monocytes

Vaccines stimulate adaptive immunity.

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# Pediatrics

Pediatrics focuses on medical care for infants and children.

## **Common Pediatric Conditions**

- Ear infections
- Asthma
- Viral fever

Vaccination schedules are important in childhood healthcare.

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## **Geriatrics**

Geriatrics focuses on healthcare for elderly individuals.

### **Common Issues**

- Dementia
- Arthritis
- Falls
- Osteoporosis

Older adults often require multidisciplinary healthcare.

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## **Dermatology**

### **Eczema**

Eczema causes itchy and inflamed skin.

#### **Symptoms**

- Dry skin
- Rash
- Itching

#### **Treatment**

- Moisturizers
  - Steroid creams
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### **Psoriasis**

Psoriasis accelerates skin cell production causing scaling.

## Symptoms

- Red patches
  - Silvery scales
  - Itching
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# Oncology

## Cancer

Cancer involves uncontrolled growth of abnormal cells.

### Common Types

- Lung cancer
- Breast cancer
- Prostate cancer
- Colon cancer

### Treatments

- Chemotherapy
- Radiation therapy
- Surgery
- Immunotherapy

Early detection improves survival rates.

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# Medical Ethics

Medical ethics includes:

- patient confidentiality
- informed consent
- beneficence
- non-maleficence

Healthcare providers must prioritize patient welfare.

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# Public Health

Public health focuses on disease prevention and community health improvement.

## Public Health Measures

- Vaccination programs
  - Clean water supply
  - Sanitation
  - Health education
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## Conclusion

Healthcare involves prevention, diagnosis, treatment, rehabilitation, and patient education. Advances in medical technology, AI-assisted diagnostics, and evidence-based medicine continue improving healthcare outcomes worldwide.